Cubism-Like Effect

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```
clear; clc; close all;
```

```
f = imread('4.2.07.tiff');
% f = imread('flowers.tif');
% f = imread('obama.jpg');
figure
imshow(f);
```



```
f = double(f);
m = size(f,1)
```

```
m = 512
```

```
n = size(f,2)
```

```
n = 512
```

```

fblur = imgaussfilt(f,0.5);
figure
imshow(uint8(fblur))
```



```
eR = edge(fblur(:,:,1), 'sobel');
eG = edge(fblur(:,:,2), 'sobel');
eB = edge(fblur(:,:,3), 'sobel');

edges = eR | eG | eB;

figure
imshow(edges)
```



```
locationArray = find(edges);
[edgeX, edgeY] = ind2sub(size(fblur), locationArray);
```

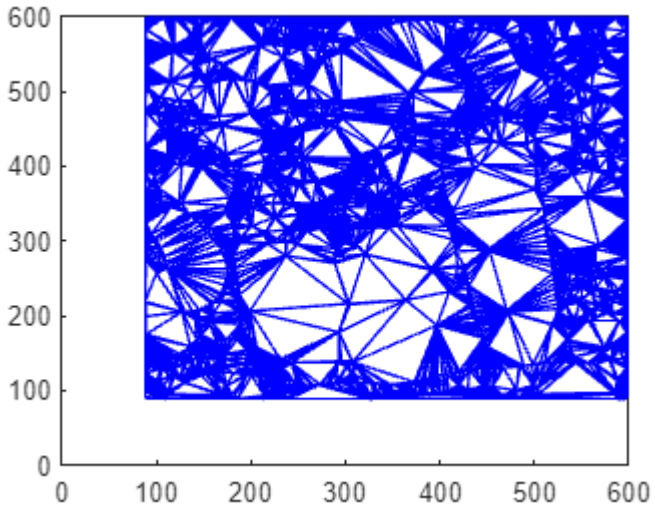
```
edgeSet = [edgeX, edgeY];

corners = [1 1; 1 n; m 1; m n];

edgeSet = cat(1, edgeSet, corners);

edgeX = edgeSet(:,1);
edgeY = edgeSet(:,2);

TRI = delaunay(edgeSet);
figure
tri = triplot(TRI,edgeX,edgeY);
rotate(tri, [0 0 1], 90);
% rotate(p, [0 1 0], 180);
rotate(tri, [1 0 0], 180);
```




```
vertex = zeros(3,2);
Tricolor = zeros(length(TRI),3);
for i = 1:size(TRI)
    vertex(1,:) = edgeSet(TRI(i,1),:);
    vertex(2,:) = edgeSet(TRI(i,2),:);
    vertex(3,:) = edgeSet(TRI(i,3),:);
    vertex = round(vertex);
    centroid = round(mean(vertex));
    pointColor = [f(centroid(1),centroid(2),1),
f(centroid(1),centroid(2),2),f(centroid(1),centroid(2),3)];
    Tricolor(i,:) = pointColor/255;
end

figure
p = patch('Faces',TRI,'Vertices',[edgeSet(:,1),-
edgeSet(:,2)],'FaceVertexCData',Tricolor,'FaceColor','flat','EdgeColor','None');
rotate(p, [0 0 1], 90);
```

```
% rotate(p, [0 1 0], 180);
rotate(p, [1 0 0], 180);
axis off
```

